



Course Name: Probability & Data Analysis

Project : #Project I

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Level: Semester I

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Here the link of form :

<https://forms.gle/3QA2BfE15gQ4VwV47>

1-Introduction:

Travel References

This survey is all about the travel experiences that have left a lasting impression on you. We're eager to learn about the destinations, books, websites, or recommendations that have inspired your journeys. Your input will help us discover valuable travel resources and ideas to share with fellow adventurers. Thank you for sharing your travel references!.



Travel Preferences Survey

Thank you for participating in our Travel Preferences Survey! We're interested in understanding your travel habits and preferences. Your feedback will help us better cater to your travel needs and improve our services. This survey consists of 10 questions and should take just a few minutes to complete. Your responses are confidential and greatly appreciated. Let's get started!

Full Name *

Short answer text

Email Address *

Short answer text

2-Data Collection:

Title: (Travel References Survey Results)

Introduction:

This dataset presents the aggregated results of the Travel References Survey, which was conducted to understand respondents' travel preferences and habits. The survey collected responses from 44 participants who shared their insights on various aspects of their travel experiences.

Data Source:

The data is derived from the responses provided by participants who completed the Travel References Survey.

Data Type: The dataset includes categorical data, as it categorizes responses into different options and calculates the percentage of respondents selecting each option.

Parameters or Variables:

The dataset includes the following parameters:

1. Travel Type Preference :

- Name: Travel Type Preference
- Type: Categorical
- Description: Indicates the type of travel preferred by respondents.
- Options: Adventure, Relaxation, Cultural, Nature, Urban

2. Frequency of Travel :

- Name: Frequency of Travel
- Type: Categorical
- Description: Indicates how often respondents travel in a year.
- Options: Rarely (0-1 times), Occasionally (2-3 times), Regularly (4-6 times), Frequently (7 or more times)

3. Traveling Companions :

- Name: Traveling Companions
- Type: Categorical
- Description: Reflects the preference for traveling alone or with others.

- Options: Solo, With friends, With family, It varies depending on the trip

4. Preferred Travel Transportation :

- Name: Preferred Travel Transportation
- Type: Categorical
- Description: Indicates the preferred mode of transportation for travel.
- Options: Car, Train, Airplane, Bus

5. Accommodation Preference :

- Name: Accommodation Preference
- Type: Categorical
- Description: Reflects the type of accommodation preferred by respondents.
- Options: Hotels, Airbnb, Hostels, etc.

6. Factors Influencing Travel Destinations :

- Name: Factors Influencing Travel Destinations
- Type: Categorical
- Description: Lists factors that influence respondents' choice of travel destinations.

- Options: Cost, Culture, Nature, Recommendations, Adventure opportunities

7. Travel Planning Style :

- Name: Travel Planning Style
- Type: Categorical
- Description: Describes respondents' approach to planning trips.
- Options: Spontaneous, Well-Planned, A mix of both

8. Travel-Related Issues :

- Name: Travel-Related Issues
- Type: Categorical
- Description: Indicates whether respondents have experienced travel-related issues.
- Options: Yes, No

9. Travel Budget Range :

- Name: Travel Budget Range
- Type: Categorical
- Description: Reflects the preferred budget range for travel.

- Options: Budget-friendly, Moderate, High-end, It varies depending on the trip

10. Domestic vs. International Travel :

- Name: Domestic vs. International Travel
 - Type: Categorical
 - Description: Indicates the likelihood of choosing domestic or international destinations for trips.
 - Options: Domestic destinations, International destinations, It depends on the trip's purpose, I haven't traveled yet
-

Data Format :

The data is presented as percentages for each option within the categorical variables.

Data Collection Method :

The data was collected through responses to the Travel References Survey, a questionnaire-based survey.

Data Context and Use : The dataset is intended for understanding the travel preferences and habits of respondents. It can be used for market research, travel planning, and analysis of the factors influencing travel choices.

References :

N/A (This dataset is derived from responses obtained in the Travel References Survey, and no external references are associated with it.)

Additional Information :

The dataset provides insights into the preferences and behaviors of 44 survey participants regarding travel. The responses are anonymized and presented as percentages to protect participants' confidentiality.

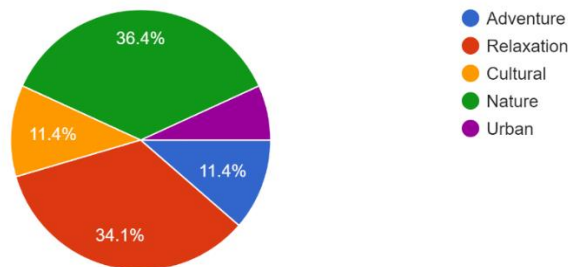
3-Data Analysis :

Categorical data & Quantitative data

1) Categorical data: Graphs & Numerical

What type of travel do you prefer?

44 responses

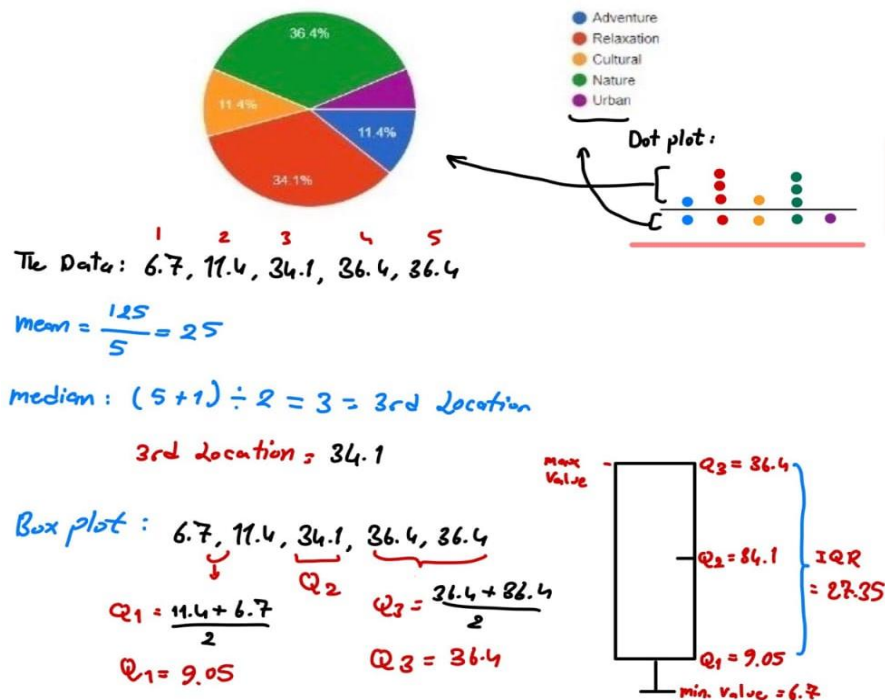


Quantitative data : Measures of Central Tendency:

What type of travel do you prefer?

44 responses

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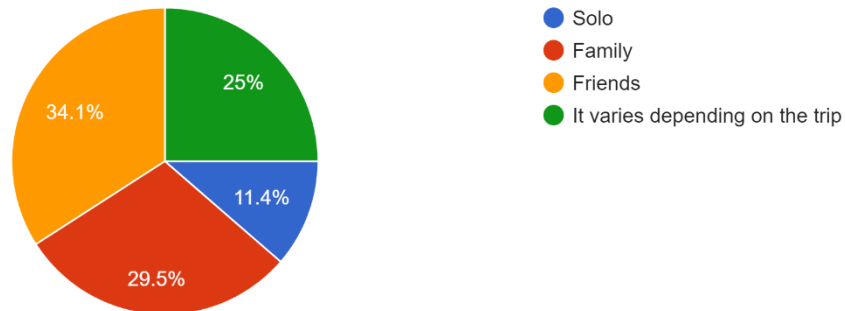


From this data we can see that most of the participants prefer to travel through nature

2) Categorical data: Graphs & Numerical

Do you prefer to travel with whom?

44 responses

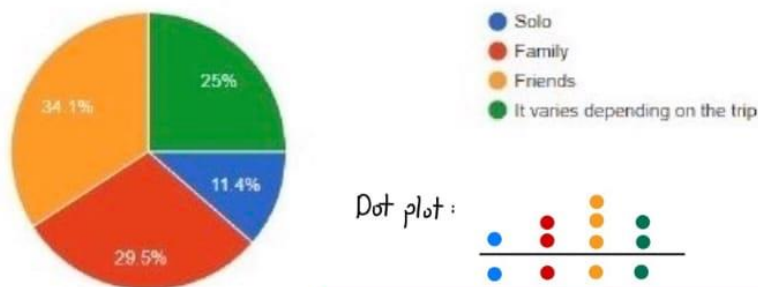


Quantitative data : Measures of Central Tendency:

Do you prefer to travel with whom?

44 responses

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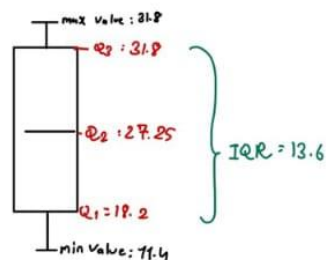


11.4, 25, 29.5, 34.1

$$\text{Mean} = \frac{100}{4} = 25$$

$$\text{Median} = \frac{25 + 29.5}{2} = 27.25$$

Box plot 11.4, 25, 29.5, 34.1
 $Q_1 = 18.2$ $Q_2 = 27.25$ $Q_3 = 31.8$

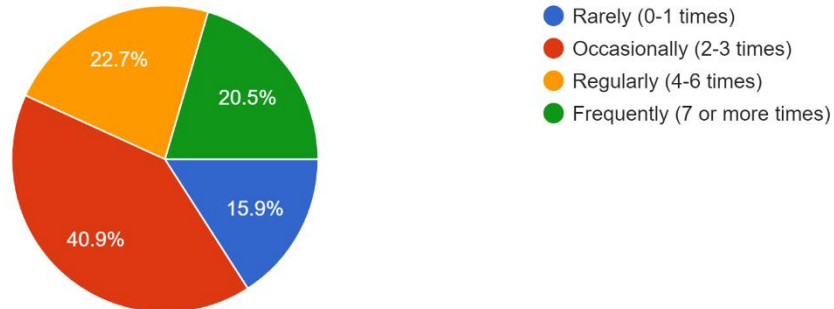


- from this data we can see that most of the participants wanted to Travel with friend's

3) Categorical data: Graphs & Numerical

How often do you travel in a year?

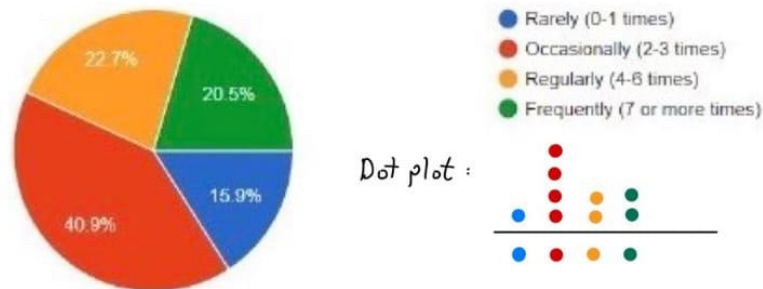
44 responses



Quantitative data : Measures of Central Tendency:

How often do you travel in a year?

44 responses



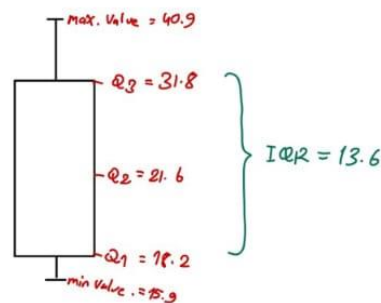
The data : 15.9 , 20.5, 22.7 , 40.9

$$\text{mean} : \frac{100}{n} = 25$$

$$\text{median} : 4 \text{ (even)}, \frac{20.5 + 22.7}{2} = 21.6$$

Box plot : 15.9, 20.5, 22.7, 40.9

$$Q_1 = 18.2, Q_2 = 21.6, Q_3 = 31.8$$

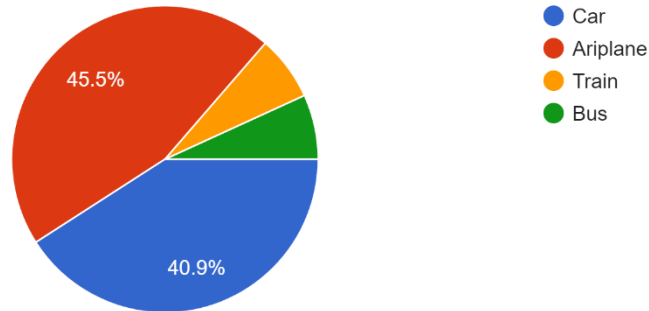


! From this data we can see that participant's had traveled 2-3 times per year on average.

4) Categorical data: Graphs & Numerical

What's your preferred mode of transportation while traveling?

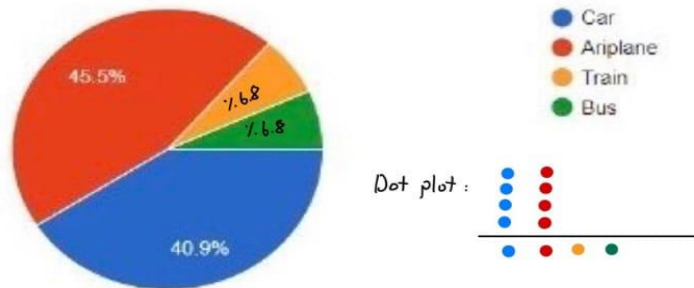
44 responses



Quantitative data : Measures of Central Tendency:

What's your preferred mode of transportation while traveling?

44 responses

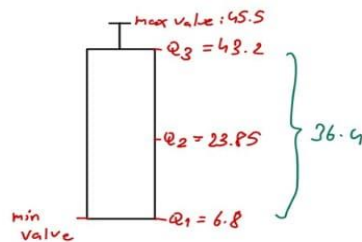


The data: 6.8, 6.8, 40.9, 45.5

$$\text{Mean} = \frac{100}{4} = 25$$

$$\text{Median} = 4 \text{ (even)} = \frac{6.8 + 40.9}{2} = 23.85$$

Box plot: 6.8, 6.8, 40.9, 45.5
 $Q_1 = 6.8$ $Q_2 = 23.85$ $Q_3 = 43.2$

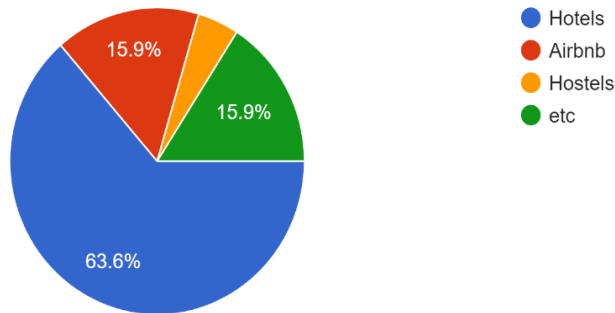


From this data of the participants that they preferred to by plane, we can observe that people's usually seek to more comfort & fast Transportation

5) Categorical data: Graphs & Numerical

What type of accommodation do you prefer?

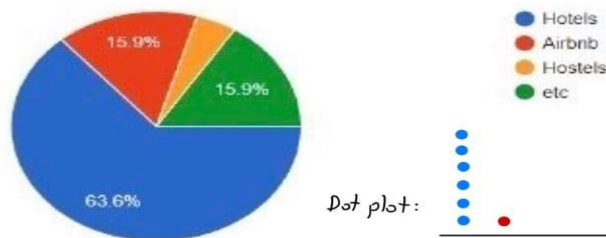
44 responses



Quantitative data : Measures of Central Tendency:

What type of accommodation do you prefer?

44 responses



The Data: 4.6, 15.9, 15.9, 63.6

$$\text{mean} = \frac{100}{4} = 25$$

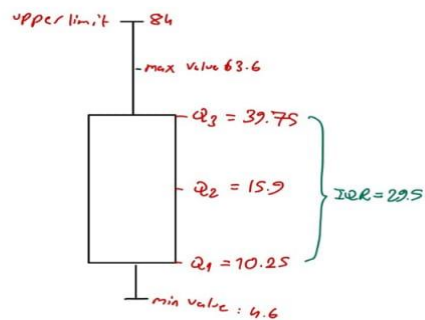
$$\text{median} = 4 \text{ (even)}, \frac{15.9 + 15.9}{2} = 15.9$$

Box plot: 4.6, 15.9, 15.9, 63.6

$$Q_1 = 10.25 \quad Q_2 = 15.9 \quad Q_3 = 39.75$$

! Since we've outlier,

$$Q_3 + 1.5 \times 29.5 \\ 39.75 + (44.25) \\ 84$$

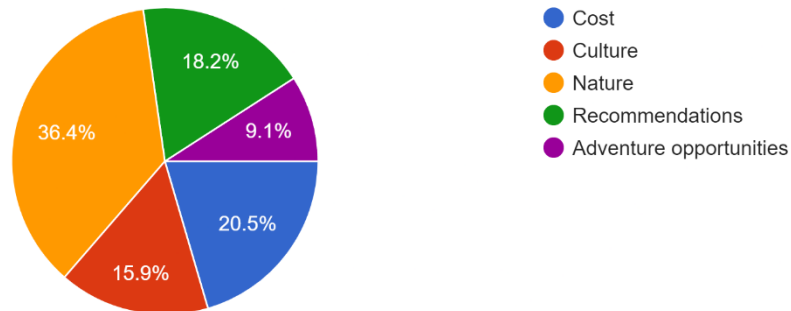


! from this Data we can observe that, majority of participant's prefer Hotels over any other accommodation

6) Categorical data: Graphs & Numerical

What factors influence your choice of travel destination?

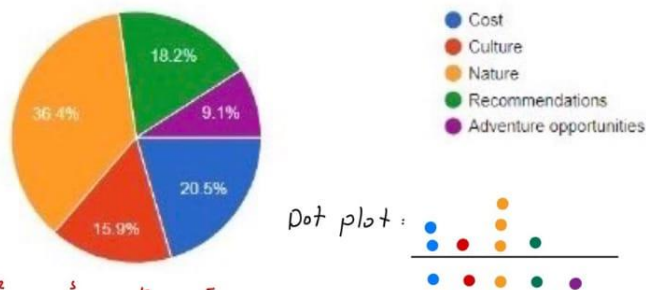
44 responses



Quantitative data : Measures of Central Tendency

What factors influence your choice of travel destination?

44 responses

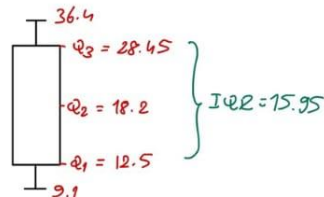


The data: 9.1, 15.9, 18.2, 20.5, 36.4

$$\text{mean} = \frac{100.1}{5} = 20.02$$

Median = 5 (odd), $(5+1) \div 2 = 3^{\text{rd}}$ location
= 18.2

Box plot: 9.1, 15.9, 18.2, 20.5, 36.4
 $Q_1 = 12.5$, $Q_2 = 18.2$, $Q_3 = 28.45$

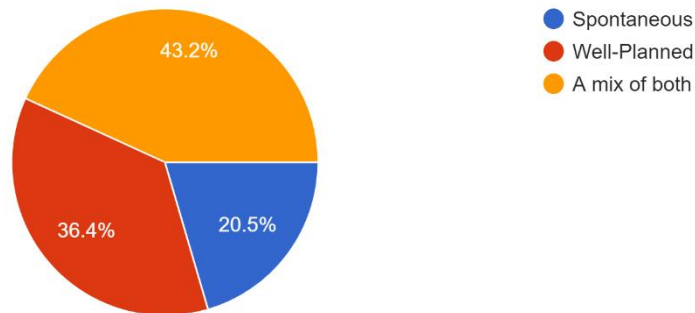


! from this data we can observe that the greatest factor that influence's participant's choice of Travel is Nature.

7) Categorical data: Graphs & Numerical

How do you plan your trips?

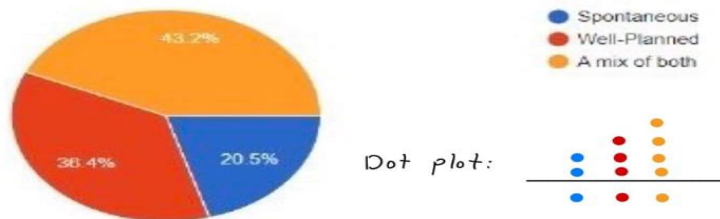
44 responses



Quantitative data : Measures of Central Tendency

How do you plan your trips?

44 responses

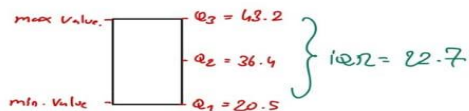


The Data: 20.5, 36.4, 43.2

$$\text{mean} = \frac{100}{3} = 33.3\%$$

$$\text{median} = 3 \text{ (odd)}, (3+1) \div 2 = 2 \text{ (2nd location)} = 36.4$$

Box plot: 20.5, 36.4, 43.2
 $Q_1 = 20.5$, $Q_2 = 36.4$, $Q_3 = 43.2$

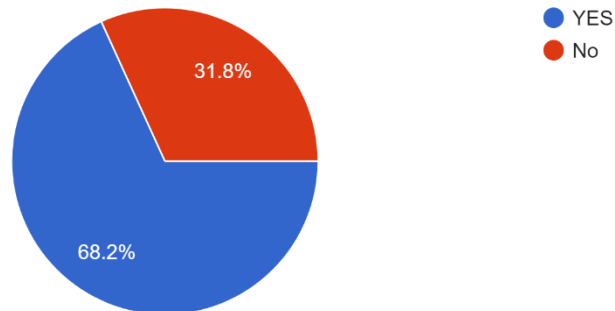


! from this data we can find-out that usually people don't plan their Travels well nor spontaneously, rather a mix of Both.

8) Categorical data: Graphs & Numerical

Have you ever experienced travel-related issues, like lost luggage or delayed flights?

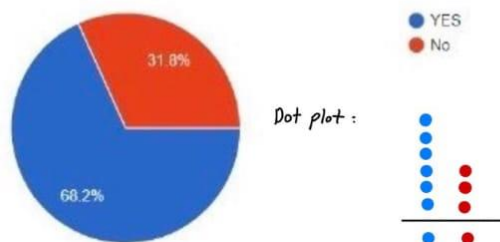
44 responses



Quantitative data : Measures of Central Tendency

Have you ever experienced travel-related issues, like lost luggage or delayed flights?

44 responses

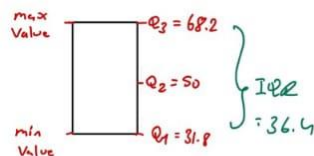


The data : 31.8 , 68.2

$$\text{mean} : \frac{100}{2} = 50$$

$$\text{median} : 2(\text{even}) = \frac{31.8 + 68.2}{2} = 50$$

Box plot : 31.8 , 68.2
 $Q_1 = 31.8$ $Q_2 = 50$ $Q_3 = 68.2$

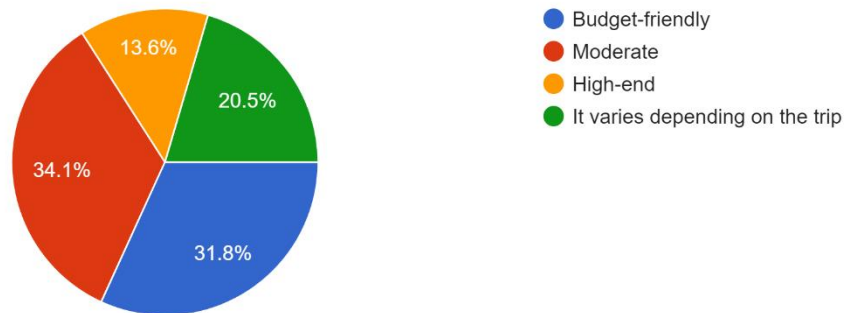


! as we look closer to this data, nearly 3 quarters of the participants had Travel-related issues.

9) Categorical data: Graphs & Numerical

What's your preferred travel budget range?

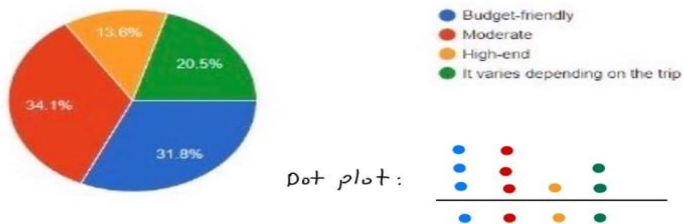
44 responses



Quantitative data : Measures of Central Tendency

What's your preferred travel budget range?

44 responses

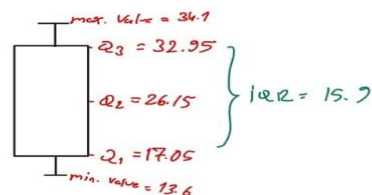


The data : 13.6, 20.5, 31.8, 34.1

$$\text{mean} = \frac{100}{n} = 25$$

$$\text{median} : n(\text{even}), \frac{20.5 + 31.8}{2} = 26.15$$

Box plot: 13.6, 20.5, 31.8, 34.1
 $Q_1 = 17.05$ $Q_2 = 26.15$ $Q_3 = 32.95$

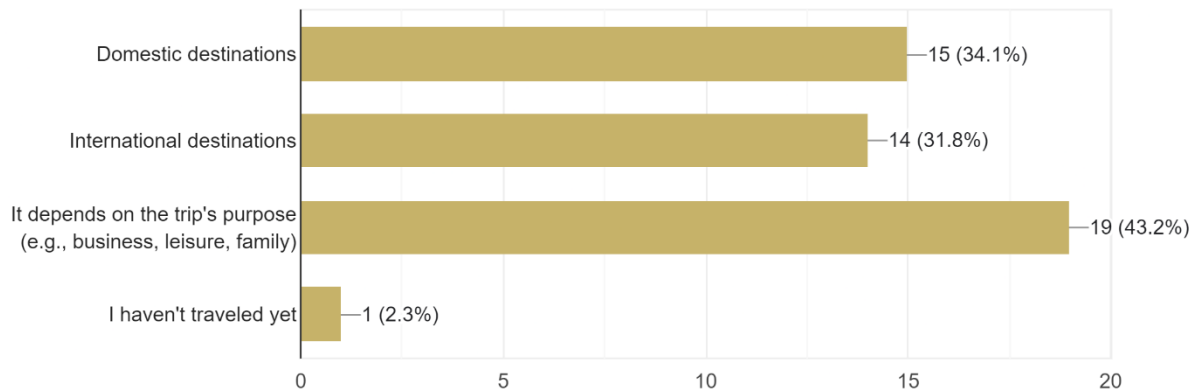


! from this data we can see that half of the participant prefer (Budget-friendly) & (moderate), while others say it varies, depending on the Trip, and few prefer (High-end) Budget.

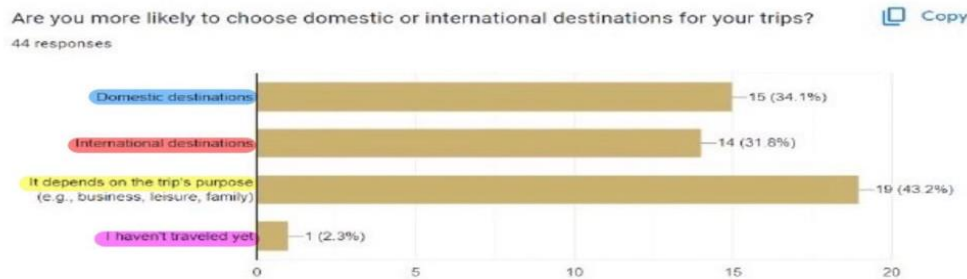
10) Categorical data: Graphs & Numerical

Are you more likely to choose domestic or international destinations for your trips?

44 responses



Quantitative data : Measures of Central Tendency

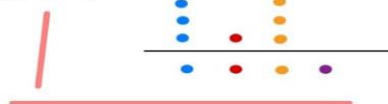


The data: 2.3, 31.8, 34.1, 43.2

$$\text{mean} = \frac{100}{4} = 25$$

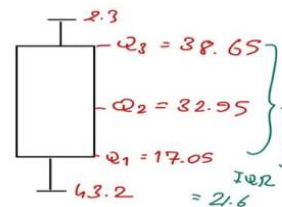
$$\text{median} = 4 \text{ (even)}, \frac{31.8 + 34.1}{2} = 32.95$$

Dot plot:



Box plot = 2.3, 31.8, 34.1, 43.2

$$Q_1 = 17.05, Q_2 = 32.95, Q_3 = 38.65$$



! from this data we can observe that,

participant's had chosen their trip destination depending on their purpose of the trip, while others think differently.

4-Conclusion :

Based on the survey responses and the provided percentage breakdowns, here's a conclusion summarizing the key findings from your "Travel References" survey:

Conclusion: The "Travel References" survey, based on 44 responses, offers valuable insights into the travel preferences and habits of participants. The results reveal several key trends and patterns that can help inform travel planning and marketing strategies:

1. Travel Type Preferences:

- Nature (36.4%) and relaxation (34.1%) are the top two preferred travel types, suggesting that respondents often seek tranquil and natural experiences.

- Adventure (11.4%) and cultural travel (11.4%) also have their place, showing a taste for unique and immersive experiences.

- Urban travel is preferred by a smaller segment of respondents (6.8%).

2. Frequency of Travel:

- Respondents vary in their travel frequency, with the majority (40.9%) traveling occasionally (2-3 times a year).
- A significant portion travels rarely (15.9%), while others travel regularly (22.7%) or frequently (20.5%).

3. Travel Companions :

- Traveling with friends (34.1%) is a popular choice, closely followed by traveling with family (29.5%).
- A notable portion (25%) indicates that their choice of companions varies depending on the trip.

4. Preferred Travel Transportation :

- Airplane travel (45.4%) is the preferred mode of transportation, reflecting the convenience and speed it offers.
- Car travel (40.9%) is another favored option, providing flexibility and control over the journey.
- Train and bus travel are less popular choices.

5. Accommodation Preferences :

- Hotels (63.6%) are the overwhelmingly preferred accommodation type, offering comfort and amenities.
- Airbnb and "other" accommodation options (both at 15.9%) also have their share of preference.
- Hostels (4.5%) are a less common choice.

6. Factors Influencing Travel Destinations :

Nature (36.4%) and recommendations (18.2%) significantly influence the choice of travel destinations, suggesting a strong inclination towards natural beauty and trusted advice.

Cost (20.5%) and culture (15.9%) are also key determinants.

Adventure opportunities (9.1%) play a smaller but noteworthy role.

7. Travel Planning Style :

Respondents' travel planning style varies, with a notable proportion (43.2%) indicating a mix of spontaneous and well-planned trips.

Well-planned trips (36.4%) are favored over completely spontaneous ones (20.5%).

8. Travel-Related Issues :

The majority of respondents (68.2%) have experienced travel-related issues like lost luggage or delayed flights, highlighting the common challenges faced during travel.

9. Preferred Travel Budget Range :

Moderate (34.1%) and budget-friendly (31.8%) travel budget ranges are the top preferences.

Some respondents (20.5%) indicate that their budget varies depending on the trip, showing flexibility in spending.

10. Domestic vs. International Travel :

- Respondents are divided in their preference for domestic destinations (34.1%) and international destinations (31.8%).
- A significant portion (43.2%) considers the trip's purpose when deciding between domestic and international travel, indicating a balanced approach.
- A small percentage (2.3%) haven't traveled yet.

These findings provide a comprehensive view of the diverse travel preferences and behaviors of the survey participants. Travel industry professionals and marketers can use this information to better cater to the needs and preferences of travelers, ensuring a more personalized and enjoyable travel experience.